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The Editor-in-Chief
Acta Astronautica
International Academy of Astronautics

Subject: Submission of original research article — *"Eclipse-Modulated Transient Energy Pumping in Near-Polar Low Earth Orbit for Spin-Stabilized Solar Sails"*

Dear Editor,

We are pleased to submit our manuscript, *"Eclipse-Modulated Transient Energy Pumping in Near-Polar Low Earth Orbit for Spin-Stabilized Solar Sails"*, for consideration as an original research article in *Acta Astronautica*. The work falls within the journal's scope of astrodynamics and space-borne system design, and addresses a question of direct relevance to passively oriented small solar-sail spacecraft in low Earth orbit (LEO).

Aim and main findings. The paper investigates whether a passively oriented, spin-stabilised solar sail in near-polar LEO can exhibit a bounded interval of net orbital-energy gain through eclipse-modulated solar-radiation-pressure (SRP) forcing. Combining a reduced analytical model with three-dimensional numerical propagations, we show that the regime exists, but only within a narrow LTDN–pitch envelope: at 700 km, only 89 of 900 sampled cases ($\sim 10\%$) show positive semi-major-axis excursion within a 14-day window, with peak gain $\Delta a \approx +20.07$ km. Eclipse is causally decisive — the eclipse-on case rises by $+10.31$ km over 14 days while the eclipse-off counterpart falls by -47.75 km — and the effect is transient and self-limiting through coupled apogee growth, eccentricity build-up, and perigee erosion.

Fit, novelty, and implications. *Acta Astronautica* publishes work in astrodynamics, space transportation, and spacecraft system design — the natural home for this contribution. To our knowledge, the eclipse-modulated transient energy-pumping regime has not previously been isolated and characterised in this form for spin-stabilised sails in LEO. Prior literature emphasises long-term or equilibrium-like SRP–drag balance; we instead show that eclipse timing alone, without active control, can flip the sign of the orbit-averaged SRP work integral over a bounded horizon, while also exposing the coupled mechanisms that close the regime. The result clarifies a real but narrow opportunity for passive orbit shaping and provides a cautionary example of how the same dynamics that enable early gain drive its eventual breakdown.

Submission status. The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All listed authors have read and approved the submitted version. We have no prior or concurrent submissions related to this work.

We thank you for considering our manuscript and look forward to the review process.

Yours sincerely,

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